

EE7207: ELECTRICAL INSTALLATIONS I
ASSIGNMENT

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Electrical System Design Concept

The design concept has been developed for a multi-occupancy residential apartment building with a Ground Floor, three typical upper floors (First Floor to Third Floor) and a Roof Floor. In high density urban residential development, the electrical distribution system should be designed to optimise energy use, human safety and continuity of supply for critical services while being fully compliant with BS 7671 (IET Wiring Regulations, 18th Edition) and local utility requirements laid down by the Ceylon Electricity Board (CEB) Sri Lanka.

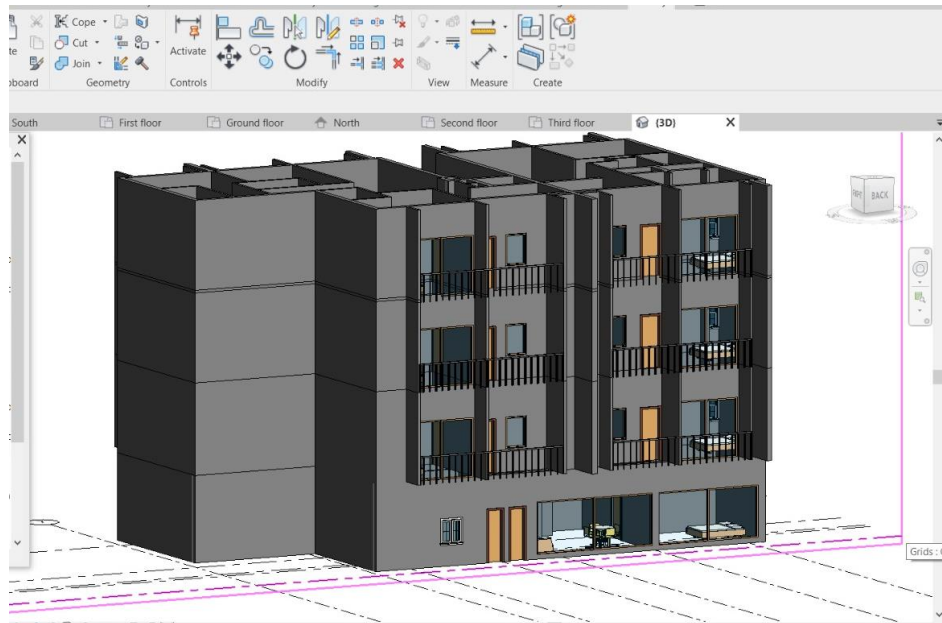


Figure 1: 3D view of the apartments from Autodesk Revit software

Table 1 : Summary of Load distribution in the apartment building

Loads	Ground Floor	Typical Upper Floor (First to Third)	Roof Floor
Occupancy Units	Lobby Area & 1 Apartment	1 Apartment per Floor (3 total)	Communal Deck & Pool
Bedrooms per Unit	1 Bedroom in GF Apartment	2 Bedrooms per Typical Apartment	Not Applicable
Air Conditioning	1 Lobby AC, 2 Apartment ACs	1 AC per Bedroom (2 ACs per Floor)	Not Applicable
Kitchen Exhaust Fans	1 Fan (0.75 kW, Single Phase)	1 Fan per Apartment (0.75 kW)	1 Central System (Part of 5 kW Load)
Pumps & Motors	2 Water Transfer Pumps (1.5 kW, 3-Ph)	Not Applicable	Pool Filtration Unit (3.0 kW, 3-Ph)
Communal Amenities	Main Entrance & Control Room	Service Risers & Elevator Core	Commercial Kitchen (5.0 kW, 3-Ph)

A standard structural and occupancy profile has been modelled to define a mathematically sound basis for all load calculations and distribution layouts. Since spatial configurations dictate local load centers, the architectural assumptions for the building have been formalized in Table 1 : Summary of Load distribution in the apartment building

1.1 Sizing and Selection of the Standby Power System

To balance structural safety with the economic requirements of the employer, backup power is restricted to the essential services of the building. These essential services are classified as,

- All fire-related and life safety systems, including fire alarm panels, emergency lighting central battery systems, and booster pumps.
- The complete electrical load of the Ground Floor and the Roof Floor.
- All lighting and socket outlets within exactly one designated bedroom of each apartment from the First Floor to the Third Floor.
- Two dedicated socket outlets in the pantry area of each typical apartment.

Uninterruptible Power Supplies (UPS) are required to supply the immediate and transient-free power for low current electronic equipment like fire alarm control panels, surveillance systems and emergency exit signage, but are technically and economically unrealistic for maintaining extended loads such as large thermal and inductive loads during a prolonged power outage. As such, a centrally located diesel generator set, with automatic mains failure (AMF) controller and 4-pole automatic transfer switch (ATS), is chosen to be the main back-up power source.

To size the generator set, the individual active power and reactive power components of the essential loads must be aggregated. So assumed power requirements are shown in the table Table 2 : Assumed power requirement of the essential loads

Standard 18,000 BTU (1.5 Ton) split AC units are assumed to have a high-efficiency inverter-driven design, drawing an electrical input active power of 1.7 kW with an operating power factor of 0.85.

The three-phase 1.5 kW water transfer pumps and the 3.0 kW pool filtration motor are assumed to operate with an power factor of 0.80 under full-load conditions.

$$apparent\ power = \frac{Real\ power}{power\ factor}$$

Table 2 : Assumed power requirement of the essential loads

Essential Load	Connected Active Power (kW)	Operating Power Factor	Diversity factor	Apparent Power (kVA)
Lobby Inverter AC (1 No.)	1.70	0.85	0.8	1.6
Ground Floor Apartment ACs (2 Nos.)	3.40	0.85	0.8	3.2
Ground Floor Kitchen Exhaust Fan	0.75	0.80	0.9	0.84375
Water Transfer Pump (1 No. Duty)	1.83	0.80	0.75	1.715625
Ground Floor Lighting & Common Sockets	4.50	0.90	0.66	3.3
Roof Floor Commercial Kitchen	5.00	1.00	0.75	3.75
Roof Floor Pool Filtration Equipment	3.0	0.80	0.8	3
Roof Floor General Power & Lighting	1.50	0.90	0.66	1.1
Essential Bedroom Loads (3 Apts)	3.00	0.90	0.8	2.66666667
Pantry Backup Sockets (3 Apts)	4.50	0.85	0.6	3.17647059
Continuous Fire & Life Safety Systems	5.00	0.80	0.7	4.375
Total Connected Essential Load	34.84kW			28.72kVA

$$\text{Total aparent power} = 28.72\text{kVA}$$

When a motor starts across-the-line (Direct-On-Line), it draws a locked-rotor current of 6 to 7 times its full-load running current. The largest motor load on the essential bus is the 3.0 kW three-phase pool pump, which has a running apparent power of 3 kVA. Sized for DOL starting, the motor exhibits an inrush starting apparent power of,

$$\text{start power} = 3\text{kVA} \times 6$$

$$\text{start power} = 18\text{kVA}$$

So, The required generator capacity can be calculated as follows

$$\text{start power} = 28.72\text{kVA} - 3\text{kVA} + 18\text{kVA}$$

$$= 41.25 \text{ kVA}$$

To ensure a conservative operating margin and accommodate future load growth, a standard 45 kVA 400V, three-phase, 50 Hz diesel generator set is selected.

1.2 Overall electrical distribution system design concept

Electrical power will be supplied to the apartment block primarily from the local electrical grid, through an outdoor substation, or an indoor substation, which will have an outdoor step-down transformer connected to the local electrical grid by the Ceylon Electricity Board (CEB). The transformer will reduce the medium voltage distribution network of the utility down to a standard low voltage (LV) distribution network, 400V/230V, 3-phase, 50 Hz.

The low voltage (LV) side of the CEB transformer will be fed by heavy duty multi-core, underground armored cables (usually copper XLPE/PVC/SWA) directly to the Main Distribution Board (MDB) which is situated in the main electrical switch room on the ground level. This point of common coupling will be the location of main utility metering, which will be installed in accordance with CEB regulations and guidelines.

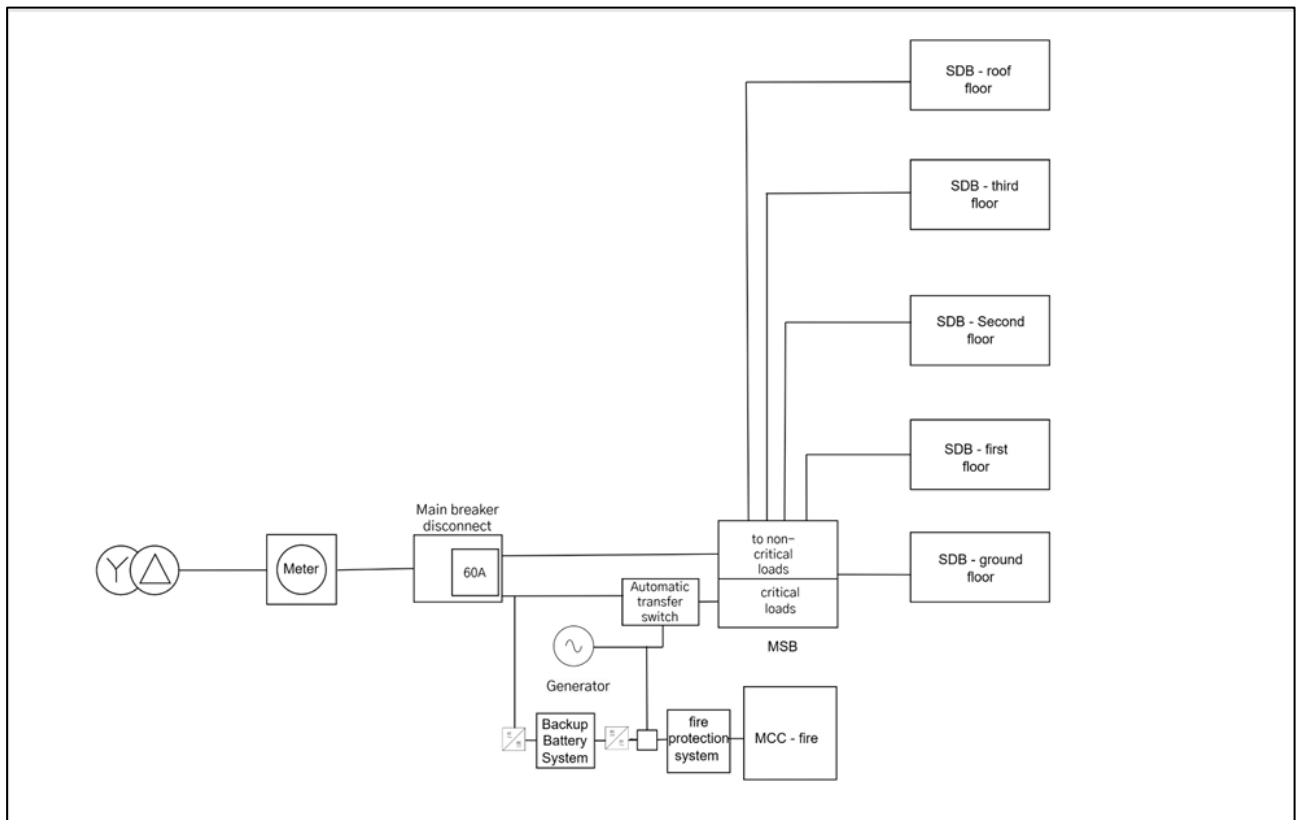


Figure.2: Overall electrical distribution system design

As shown in the figure 1, For vertical and horizontal main power distribution throughout the building, a robust Cable Tray and Cable Ladder System will be utilized.

A cable tray system is chosen for the electrical distribution in the proposed 5-story apartment building for techno-economic reasons, rather than a busbar trucking system. The high installation and material costs of busbar systems are not justifiable in terms of the economy of the building, as the building height and load are relatively moderate. Rather, a perforated hot dip galvanized (GI) cable tray

assembly incorporating XLPE insulated multi-cores offer a cost effective, versatile and dependable solution for power distribution. The vertical power distribution will be done via cable ladder through a dedicated electrical riser duct and the controls and distribution boards of the apartments will be connected through the standard 10-20-amp electrical distribution boxes with a standard cable joint, providing ease in installation, maintenance and future changes in use applicable to the scale of the apartment building.

The electrical distribution system consists of two busbars on the Main Distribution Board (MDB) that operate in parallel to separate essential and non-essential electrical loads to enable efficient management of backup power. An Automatic Transfer Switch (ATS) panel is used to switch between the utility service, standby generator and essential busbar. The ATS automatically switches critical loads to the generator during a power failure, once the generator becomes stable.

Backup loads include all Ground Floor and Roof Floor services including the lobby air conditioning system, water pumps, pool equipment and Kitchen loads on the roof. Furthermore, apartment loads are backed up, such as lighting circuits and socket outlets of one of the bedrooms in each apartment, and two pantry socket outlets for each apartment from the 1st to 3rd floor.

Non-essential loads are not backed-up; they are connected to a utility-only MDB section. These consist of general purpose socket outlets and lighting circuits, kitchen exhaust fans, and apartment air conditioning units. This separation reduces the need for generator capacity and optimizes fuel economy when operating in emergency mode.

1.3 Selection of Lighting Fittings

Lighting design is based on achieving an average illuminance of approximately 300 lux in general living areas, which is suitable for residential apartment lighting according to recommended standards. The design concept incorporates the following factors in compliance with BS 7671 standards:

- ❖ **Illuminance Level:** Target illuminance is set around 300 lux for living spaces to ensure adequate visibility and comfort.
- ❖ **Lighting Uniformity:** Uniform distribution of light to avoid glare and shadows, enhancing visual comfort.
- ❖ **Color Temperature:** Use of LED fittings with color temperatures between 3000K (warm white) and 4000K (neutral white) to create a comfortable and natural ambiance.

Selection of lighting fittings are as follows.

Table 3: Selected Lighting Fittings for Different Building Areas

Location	Specific Room / Zone	Lux level	Selected CoreLine Model & Type	Design Purpose & Application
Ground Floor	Lobby & Reception Area	250	CoreLine Downlight (DN140B)	provide a warm, low-glare, and welcoming aesthetic for visitors.
	Driver's room	200	CoreLine Panel (RC132V)	shadow-free light across multi-functional living spaces.
	Parking	200	Pacific LED Gen5	suitable for semi outdoor places
	Electric meters and panel	250	CoreLine Downlight (DN140B High-CRI)	high color rendering over countertops for safety and reading
	Water Pump Room	150	Ledinaire Waterproof	A cost-effective, Heavy-duty, dust-tight, and jet-waterproof sealed environments.
First Floor to Third Floor	Apartment Bedrooms (Each Floor)	150	CoreLine Recessed Spot (RS141B)	Low-glare recessed directional spotlights to create a calm and restful mood
	Apartment Living & Dining Areas	200	CoreLine Panel (RC132V)	General ambient illumination designed to maintain a spacious, modern feel across the main residential suites.
	Apartment Kitchen & Pantry Areas	250	CoreLine Downlight (DN140B High-CRI)	Dedicated downlighting positioned intentionally over the counter and food preparation spaces.
Common Access (All Floors)	Main Corridors & Lift Lobbies	150	CoreLine Downlight (DN140B)	Reliable, long-life recessed downlighting for clean, continuous, and efficient public circulation lighting.
	Escape Staircases & Service Shafts	150	CoreLine Wall-Mounted / Bulkhead (WL140V)	Vandal-proof (IK10) utility fixture with optional motion sensors
Roof Floor	Kitchen Area	250	CoreLine Downlight (DN140B High-CRI)	High-output, moisture-tolerant downlights designed to handle the elevated steam

	Pool Equipment & Filter Room	150	CoreLine Waterproof (WT120C)	IP66 weather-sealed linear casing engineered to protect against moisture around pool systems.
	Open Roof Common Areas & Pool Perimeter	100	CoreLine Bulkhead (WL140V IP65)	Weatherproof, UV-stabilized exterior fixtures designed to safely illuminate outdoors

1.4 Selection of protective devices and coordination philosophy

The grading and selection of protective devices should isolate the fault on any branch circuit by the nearest protective device in the service line, thereby limiting the impact of the fault condition on the remainder of the installation to ensure a safe, reliable, continuous electrical supply. The final branch circuits will use circuit breakers that meet local standards to provide reliable and economical service, and the large sub-distribution and main panels will be equipped with high performance switchgear to offer greater protection, durability and reliability.

The grading and characteristics of all circuit breakers in the building are selected based on the nature of their downstream loads to prevent nuisance tripping.

Table 4: Protective Device Selection and Coordination Characteristics

Load / Circuit Type	Protective Device	Application / Protection Characteristics
Lighting Circuits	6A Type B MCBs	The Type B magnetic trip characteristic provides sensitive protection for resistive and electronic LED lighting loads.
General Purpose Socket Outlets	20A Type B MCBs for radial circuits or 32A Type B MCBs for ring final circuits	Suitable for standard socket outlet circuits and general purpose loads.
Air Conditioning Units	16A Type C MCBs	The Type C magnetic trip characteristic accommodates inductive compressor starting inrush currents without nuisance tripping.

Water Transfer Pumps (1.5kW)	10A Type C Adjustable 4–6.3A MPCB,	Protects against short circuit, overload, and phase failure
Kitchen Exhaust Fans	6A Type C MCBs	Supports single-phase motor inrush currents effectively.
Water Transfer Pumps & Pool Equipment	10A Type C Three-Phase MCBs	Matched to the running currents of the 1.5 kW and 3.0 kW motors for reliable motor protection.

To accommodate the transient inrush currents of the 1three-phase pumps.5 kW and AC units. Type C characteristic protective devices are utilized on all inductive motor circuits to prevent nuisance tripping.

Surge Protective Devices (SPD) is a specialized component designed to protect sensitive electrical equipment, electronics, and appliances from transient overvoltage events, commonly known as voltage spikes or surges due to lightning.

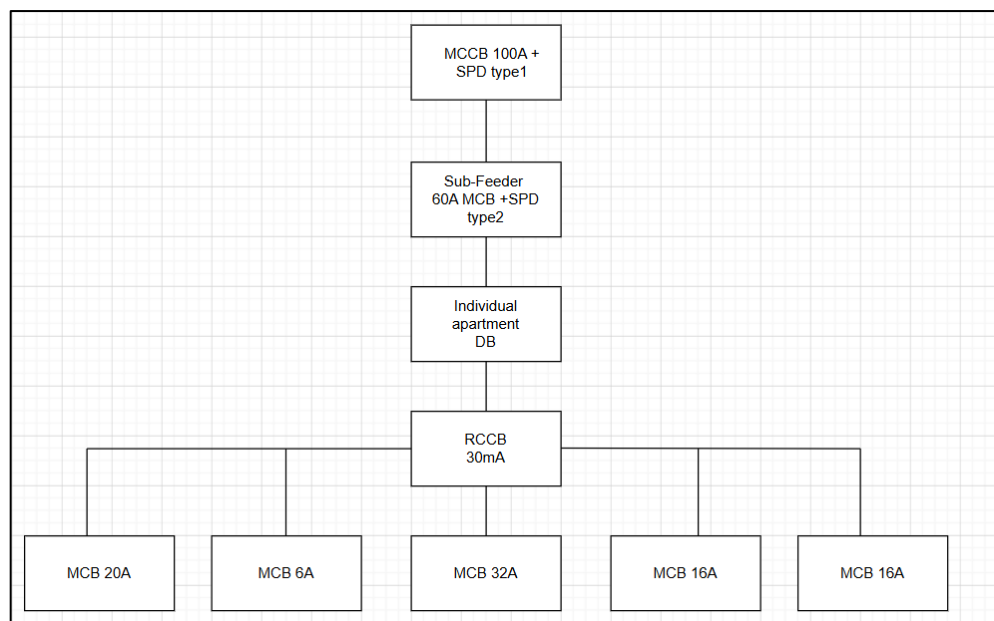


Figure 3: Coordination philosophy of protective devices

As shown in the figure 2 the overcurrent coordination scheme is designed to achieve total selectivity, ensuring that under any fault condition, the protective device closest to the fault opens first while all upstream devices remain closed.

Furthermore, a full-sized Neutral conductor is specified across all multi-core distribution feeders to safely mitigate the cumulative current risks associated with triplen harmonics from non-linear domestic appliances and LED lighting installations

2 Lighting Design

After deciding the required illumination level for each location, the total number of luminaires required for each specific space is calculated utilizing the standard Average Lumen Method formula

$$\text{Total number of luminaries} = \frac{\text{Area} \times \text{Required level of illumination}}{\text{Lumen output per luminaire} \times Z \times UF \times MF}$$

Where:

UF = Utilization Factor (Assumed as 0.6 based on standard room indices)

MF = Maintenance Factor (Assumed as 0.8 for this design)

Z = Number per lamp for each luminary

2.1 Lighting Design for the Ground floor and Roof floor common areas.

Specimen Calculation for the ground floor common areas are as follows,

Lobby area in the ground floor,

$$\begin{aligned}\text{Required lux level} &= 150 \\ \text{Lobby area} &= 2.095062 \text{ m}^2 \\ \text{Lumen output per Luminary} &= 2200 \\ \text{Maintenance Factor} &= 0.8 \\ \text{Utilization Factor} &= 0.6 \\ \text{Number per lamp for each luminary} &= 1 \\ \text{Total number of luminaries} &= \frac{28.82506 \times 150}{2200 \times 1 \times 0.6 \times 0.8} \\ &= 4.0944\end{aligned}$$

Specimen Calculation for roof floor common areas (Gym in the roof floor),

$$\begin{aligned}\text{Required lux level} &= 300 \\ \text{Lobby area} &= 13.69 \text{ m}^2 \\ \text{Lumen output per Luminary} &= 3600 \\ \text{Maintenance Factor} &= 0.8 \\ \text{Utilization Factor} &= 0.6 \\ \text{Number per lamp for each luminary} &= 1 \\ \text{Total number of luminaries} &= \frac{13.69 \times 300}{3600 \times 1 \times 0.6 \times 0.8} \\ &= 2.376\end{aligned}$$

Table 5: Lighting Design Calculations for Ground Floor and Roof Floor Areas

Floor	Calculated Space	Calculation (L×W)	Area (m2)	Lux Level (E)	Lumen Output per luminaire	Z	UF	MF	Calculated	No. of luminaries
Ground	Bed Room	(4.50×3.525) + (1.8×2.225)	19.87	100	1100	1.0	0.6	0.8	3.76	4
	Dining and Living Room	5.668×2.9	16.44	200	2100	1.0	0.6	0.8	3.261	4
	Pantry	2.025×2.975	6.02	250	1100	1.0	0.6	0.8	2.85	3
	Toilet	2.225×1.700	3.78	150	1100	1.0	0.6	0.8	1.07	1
	Driver's Room	2.667×1.800	4.80	200	1100	1.0	0.6	0.8	1.808	2
	Care Taker Room	2.50×1.725	4.31	150	1100	1.0	0.6	0.8	1.22	1
	Showers Room	1.5×0.9	1.35	150	1100	1.0	0.6	0.8	0.38	1
	Driver's Toilet	1.5×0.9	1.35	150	1100	1.0	0.6	0.8	0.38	1
	Pump Room	4.914×2.0+1.8×0.3	10.37	150	4000	1.0	0.6	0.8	0.81	1
	Elect. Meters & Panels	1.300×2.400	3.12	200	2500	1.0	0.6	0.8	0.52	1
	Garbage Room	1.300×2.400	3.12	100	1350	1.0	0.6	0.8	0.48	1
	GF Lobby & Stairs	(3.926×3.537) + (2.400×6.337)	29.10	150	2200	1.0	0.6	0.8	4.13	4
	Parking Area	(2.40×4.80) ×3 x 2	69.12	100	4000	1.0	0.6	0.8	1.8	4
Roof Top	Generator Room	2.50×3.300	8.25	150	4000	1.0	0.6	0.8	0.64	2
	gym	4.10×3.338	13.69	300	3600	1.0	0.6	0.8	2.38	4

	Changing Room & Toilet	1.50×0.900	1.35	150	1100	1.0	0.6	0.8	0.38	1
	Bar / BBQ Counter	3.113×2.039	6.35	150	1350	1.0	0.6	0.8	1.47	2
	Roof Lobby & Stairs	2.400×6.337	15.21	150	2200	1.0	0.6	0.8	2.16	2
	Open Terrace	7.325×6.337 + 18.62	65.040	100	2400	1	0.6	0.8	5.64	6
	corridor	1.325x7.960	10.547	150	2200	1	0.6	0.8	1.498	2
	pool deck	3.4x11.038 + 3.5x3.5	49.779	100	1000	1	0.6	0.8	10.37	10
	pool	3300x7338	24.215	200	1500	1	0.6	0.8	6.726	6
	Lift machine room	2.625 x 4.362	11.45	150	1100	1	0.6	0.8	3.25	4

2.2 Lighting Design for the Apartments.

Specimen Calculation for master Bed room in apartments

$$\text{Required lux level} = 100$$

$$\text{Lobby area} = 14.04 \text{ m}^2$$

$$\text{Lumen output per Luminary} = 1100$$

$$\text{Maintenance Factor} = 0.8$$

$$\text{Utilization Factor} = 0.6$$

$$\text{Number per lamp for each luminary} = 1$$

$$\begin{aligned} \text{Total number of luminaries} &= \frac{14.04 \times 100}{1100 \times 1 \times 0.6 \times 0.8} \\ &= 2.659 \end{aligned}$$

Table 6: Lighting Design Calculations for Typical Apartment Floors

Floor	Calculated Space	Calculation (L×W)	Area (m2)	Lux Level (E)	Lumen Output per Luminaire (Φ)	Z	UF	MF	Raw Calculated (Nraw)	Selected (Ndesign)
1-3 left Right side	Master Bedroom L	3.900×3.600	14.04	100	1100	1.0	0.6	0.8	2.66	4
	Bed Room L	3.900×3.300	12.87	100	1100	1.0	0.6	0.8	2.44	3
	Sitting Areas	3.80×6.337	24.08	150	2100	1.0	0.6	0.8	3.58	4
	Dining Areas	3.80×2.025	7.70	200	1100	1.0	0.6	0.8	2.19	2
	Pantries	3.80×2.400	9.12	250	1100	1.0	0.6	0.8	4.318	4
	Kitchen Yards	3.80×1.500	5.70	150	1350	1.0	0.6	0.8	1.319	1
	Master Toilets	2.70×1.50	4.05	150	1100	1.0	0.6	0.8	1.15	1
	Master Balcony	3.900×1.500	5.85	100	1350	1.0	0.6	0.8	0.9	1
	Balcony	3.90×1.00	3.90	100	1350	1.0	0.6	0.8	0.6	1
	Master Bedroom R	(3.900×3.285) +(1.475×2.700)	16.79	100	1100	1.0	0.6	0.8	3.18	4
	Bed Room 1R	3.348×3.000	10.04	100	1100	1.0	0.6	0.8	1.9	2
	Bed Room 2R	3.382×3.833	12.96	100	1100	1.0	0.6	0.8	2.46	4
	Master Toilets	2.70×1.40	3.78	150	1100	1.0	0.6	0.8	1.07	2
	Toilets 1	2.70×1.50	4.05	150	1100	1.0	0.6	0.8	1.15	1
	Toilets 2	2.70×1.40	3.78	150	1100	1.0	0.6	0.8	1.07	1
	Sitting Areas	4.005×6.337	25.38	150	2100	1.0	0.6	0.8	3.78	4
	Dining Areas	3.631×2.025	7.35	200	1100	1.0	0.6	0.8	2.77	2
	Pantries	4.20×2.40	10.08	250	1100	1.0	0.6	0.8	4.77	5
	Kitchen Yards	4.20×1.50	6.30	100	1350	1.0	0.6	0.8	0.97	1

	Master Balcony 1(DW5)	3.9×1.50	5.85	100	1350	1.0	0.6	0.8	0.9	1
	Balcony 1	3.0×1.50	4.50	100	1350	1.0	0.6	0.8	0.69	1
	Balcony 2	3.833×1.00	3.83	100	1350	1.0	0.6	0.8	0.59	1
	Typ. Lobby & Stairs	6.337×2.400	15.21	150	2200	1.0	0.6	0.8	2.16	2

I have use the Philips professional Lighting catalog and The table 7 shows the parameters of the selected luminaries extracted from the catalog [1]

Table 7: Technical Specifications of Selected Philips Lighting Fixtures

Philips Model Code / Series	Power (W)	Lumen Flux (Φ , lm)	Efficacy (lm/W)	Typical Application
DN145B LED10S/830 PSU II WH (CoreLine Slim Downlight)	11.0	1100	100	Bedrooms, Pantries, Dining Rooms
DN145B LED20S/830 PSU II WH (CoreLine Slim Downlight)	21.0	2100	100	Sitting Areas
DN140B LED10S/830 PSU C IP54 P16 (CoreLine Downlight)	9.5	1100	106	Toilets, Shower Rooms
DN140B LED20S/840 PSU C P16 (CoreLine Downlight)	19.0	2200	111	Common Lobbies, Stairs
RC132V G5 36S/840 PSD W60L60 OC (CoreLine Panel)	28.5	3600	125	Gym
WT120C G2 LED40S/840 WIA L1200 (CoreLine Waterproof)	30.0	4000	133	Generator Room, Pump Room, Parking

BN126C LED25S/840 PSU L1200 (CoreLine Batten)	18.0	2500	130	Electrical Panel Room
WL140V LED12S/840 PSU WH (CoreLine Wall-Mounted)	12.0	1350	110	Kitchen Yards, Balconies, Garbage Room, BBQ
BVP164 LED24/840 PSU 20W SWB CE (Ledinaire Floodlight)	20.0	2400	120	Open Terrace
SmartBright Direct Panel (G2)	28	2200	110	corridors
Philips Wall-Mounted LED (Outdoor)	15	1000	100	Pool deck
Philips Ledinaire Underwater/Spot	20	1500	120	Pool

3 Electrical Demand Load Calculation

3.1 Lighting load in the apartment

Table 8: Lightning load in the apartment building

Floor	Space	Type of Light	Power Per Lamp (W)	Selected No. of Luminaries
Ground floor	Bed Room	CoreLine Slim Downlight 10s	11.0	4
	Dining and Living Room	CoreLine Slim Downlight 20s	21.0	4
	Pantry	CoreLine Slim Downlight 20s	21.0	3
	Toilet	CoreLine Downlight	9.5	2
	Driver's Room	CoreLine Slim Downlight	11.0	2

	Care Taker Room	CoreLine Slim Downlight	11.0	1
	Showers Room	CoreLine Downlight	9.5	1
	Driver's Toilet	CoreLine Downlight	9.5	1
	Pump Room	CoreLine Waterproof	30	1
	Elect. Meters & Panels	CoreLine Batten	18	1
	Garbage Room	CoreLine Wall-Mounted	12	1
	GF Lobby & Stairs	CoreLine Slim Downlight	21	4
	Parking Area	CoreLine Waterproof	30	4
	Drive way	Ledinaire Floodlight	20	6
	corridor	SmartBright Direct Panel (G2)	28	2
Roof floor	Generator Room	CoreLine Waterproof	30	2
	gym	CoreLine Panel	28.5	4
	Changing Room & Toilet	CoreLine Downlight	9.5	1
	Bar / BBQ Counter	CoreLine Wall-Mounted	12	2
	Roof Lobby & Stairs	CoreLine Downlight	19	2

	Open Terrace	Ledinaire Floodlight	20	6
	corridor	SmartBright Direct Panel (G2)	28	2
	pool deck	Philips Wall-Mounted LED (Outdoor)	15	10
	pool	Philips Ledinaire Underwater/Spot	20	6
	Lift machine room	CoreLine Downlight	11	4
1 to 3 rd floor left apartment	Master Bedroom L	CoreLine Slim Downlight LED10S	11	4
	Bed Room L	CoreLine Slim Downlight LED10S	11	3
	Sitting Areas	CoreLine Slim Downlight LED20S	21	4
	Dining Areas	CoreLine Slim Downlight LED10S	11	2
	Pantries	CoreLine Slim Downlight LED10S	11	4
	Kitchen Yards	(CoreLine Wall-Mounted	12	1
	Master Toilets	CoreLine Downlight IP54 P16	9.5	2
	Master Balcony	CoreLine Wall-Mounted	12	1
	Balcony	CoreLine Wall-Mounted	12	1
1 to 3 rd floor right side apartment	Master Bedroom R	CoreLine Slim Downlight LED10S	11	4

	Bed Room 1R	CoreLine Slim Downlight LED10S	11	2
	Bed Room 2R	CoreLine Slim Downlight LED10S	11	4
	Master Toilets	CoreLine Downlight IP54 P16	9.5	2
	Toilets 1	CoreLine Downlight IP54 P16	9.5	2
	Toilets 2	CoreLine Downlight IP54 P16	9.5	2
	Sitting Areas R	CoreLine Slim Downlight LED20S	21	4
	Dining Area R	CoreLine Slim Downlight LED10S	11	2
	PantriescR	CoreLine Slim Downlight LED10S	11	5
	Kitchen Yards	CoreLine Wall- Mounted	12	1
	Master Balcony 1(DW5)	CoreLine Wall- Mounted	12	1
	Balcony 1	CoreLine Wall- Mounted	12	1
	Balcony 2	CoreLine Wall- Mounted	12	1
Common	Typ. Lobby & Stairs	CoreLine Downlight C P16	19	2

3.2 Ground Floor Demand Load Calculation

Table 9: The Diversity factor and the utilization factor for the loads

Load Type	Diversity Factor (ks)	Utilization Factor (ku)
Air Conditioning	0.8	0.9
Lighting	0.66	0.85
Socket Outlets	0.5	0.85
Electric Cookers/Ovens	0.4	0.95
Water Heaters/Showers	0.9	1.0
Pumps (Water/Sump)	0.75	0.9
Lifts / Elevators	0.7	0.9

Specimen calculation for air conditioning machine in the lobby area can be calculated using Assumptions for diversity factor and utilization factor are shown in the table

$$\begin{aligned}
 \text{Power} &= 1.7 \text{ kW} \\
 \text{No of units} &= 1 \\
 \text{Power factor} &= 0.85 \\
 \text{Total kVA rating} &= \frac{\text{Real power}}{\text{power factor}} \\
 &= \frac{1.70}{0.85} \\
 &= 2 \text{ kVA} \\
 \text{Diversity factor} &= 0.8 \\
 \text{Utilization factor} &= 1 \\
 \text{Total load with factors in kVA} &= \text{Num. of units} \times \text{Power in kVA per unit} \times k_s \times k_u \\
 &= 1 \times 2\text{kVA} \times 0.8 \times 0.9 \\
 &= 1.44 \text{ kVA} \\
 \\
 \text{Total lighting power for the} &= 11 \times 4 + 21 \times 4 + 21 \times 3 + 9.5 \times 2 \\
 \text{ground floor apartment} &= 210 \text{ W}
 \end{aligned}$$

$$\begin{aligned} \text{Total lighting power for the ground floor common area} &= 11 \times 2 + 11 \times 1 + 9.5 \times 2 + 30 \times 1 + 18 \times 1 + 12 \times 1 + 21 \times 4 \\ &= 196 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Total lighting power for the parking and drive way} &= 30 \times 4 + 20 \times 6 \\ &= 120 \text{ W} \end{aligned}$$

Table 10: Ground Floor Electrical Demand Load Calculation

Area	Space	Load	No. of Units	KW per Load	Power Factor	KVA	Diversity Factor	Utilization Factor	Total KVA with Factors
Apartment	Bedroom	AC	1	1.7	0.85	2	0.8	0.9	1.44
		Sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
	Dining Room	AC	1	1.7	0.9	2	0.8	0.9	1.44
		Sockets	5	0.15	0.9	0.18	0.5	0.9	0.405
	Kitchen	Exhaust Fan	1	0.75	0.9	0.83	1	1	0.83
		Sockets	5	0.5	0.95	0.6	0.4	1	1.2
	toilet	Sockets	2	0.15	0.85	0.18	0.5	1	0.18
	All	Lighting	13	0.21	0.9	$\frac{0.2333}{3}$	0.66	0.85	0.1309
Common	Lobby	AC	1	1.7	0.85	2	0.8	0.9	1.44
	Pump room	Water Pumps	1	1.5	0.9	1.67	0.75	0.9	$\frac{1.1272}{5}$
		Sockets	4	0.15	0.9	0.18	0.5	1	0.36

	Elect Meter Panel	Sockets	2	0.15	0.9	0.18	0.5	1	0.18
	Driver's room	Sockets	2	0.15	0.9	0.36	0.5	1	0.36
	Care Takers Room	sockets	2	0.15	0.9	0.18	0.5	1	0.18
	all lighting in	lighting	12	0.196	0.9	0.22	0.66	0.85	0.123
	Parking Area and drive way	Sockets	6	0.5	0.9	0.18	0.5	1	0.54
		Lighting	4+6	0.24	0.9	0.2667	0.66	0.85	0.1496
								Total KVA	

Total demand in ground floor in kVA = 9.5521kVA

Ground Floor with 20% expansion = 9.5521 x 1.2

= 11.46 kVA

$$\begin{aligned}
 \text{Total design current} &= \frac{\text{Total load} \times 1000}{\sqrt{3} \times 400} \\
 &= \frac{11.46 \times 1000}{\sqrt{3} \times 400} \\
 &= 16.54 \text{ A}
 \end{aligned}$$

3.3 Roof Floor Demand Load Calculation

$$\begin{aligned}
 \text{Total lighting power for the Roof floor} &= 30 \times 2 + 28.5 \times 4 + 9.5 \times 1 + 12 \times 2 + 19 \times 2 + 20 \times 6 + \\
 &\quad 28 \times 2 + 15 \times 10 \\
 &= 571.5 \text{ W}
 \end{aligned}$$

Table 11: Roof Floor Electrical Demand Load Calculation

Space	Load	No. of Units	KW per Load	Power Factor	KVA	Diversity Factor	Utilization Factor	Total KVA with Factors
Bar / BBQ Counter	Kitchen Equipment	1	5	0.9	5.5556	0.8	1	4.444
	Sockets	4	0.5	0.85	0.6	0.5	0.85	1.306
Pool and pool deck	Pool Equipment	1	3	0.9	3.33	0.9	1	3
	Sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
Generator Room	Sockets	3	0.15	0.85	0.18	0.5	0.85	0.229
Changing Room & Toilet	Sockets	2	0.15	0.85	0.18	0.5	0.85	0.153
Open Terrance	sockets	6	0.15	0.85	0.18	0.5	0.85	0.459
Elevator	motor	1	6	0.9	6.6667	0.7	0.9	4.2
Roof floor	lighting	29	0.571	0.9	0.6344	0.66	0.85	0.356
Total KVA							13.45	

Total demand in roof floor in kVA = 13.45 kVA

with 20% expansion = 13.45×1.2

= 16.14 kVA

Total design current = $\frac{\text{Total load} \times 1000}{\sqrt{3} \times 400}$

= $\frac{13.45 \times 1000}{\sqrt{3} \times 400}$

= 19.41A

3.4 1st floor to 3rd floor Demand Load Calculation

3.4.1 Deman load calculation for left side apartment

$$\begin{aligned} \text{Total lighting power for the left} &= 11 \times 7 + 21 \times 4 + 11 \times 6 + 12 \times 3 + 9.5 \times 2 \\ \text{side apartment} &= 282 \text{ W} \end{aligned}$$

Table 12: Left-Side Apartment Demand Load Calculation

Area	Space	Load	No. of Units	KW per Load	Power Factor	KVA	Diversity Factor	Utilization Factor	Total KVA with Factors
Apartment	master Bedroom	AC	1	1.7	0.85	2	0.8	0.9	1.44
		socket	4	0.15	0.85	0.18	0.5	0.85	0.306
	Bedroom	AC	1	1.7	0.85	2	0.8	0.9	1.44
		Sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
	Dining Room and sitting room	AC	1	1.7	0.85	2	0.8	0.9	1.44
		Sockets	6	0.15	0.85	0.18	0.5	0.85	0.459
	Pantry & kitchen yard	Exhaust Fan	1	0.75	0.9	0.833333	0.9	1	0.75
		Sockets	6	0.5	0.85	0.6	0.5	0.85	1.519
	Balcony	sockets	3	0.15	0.85	0.18	0.5	0.85	0.2295

	Toilets	sockets	2	0.15	0.85	0.18	0.5	0.85	0.153
	all	lighting	22	0.282	0.9	0.31333	0.66	0.85	0.17578
Common	Lobby & Stair	sockets	2	0.15	0.85	0.18	0.5	0.85	0.153
		Lighting	2	0.038	0.9	0.04222	0.66	0.85	0.02369
Total KVA							8.406		

$$\begin{aligned}
 \text{Total demand for a left side apartment in kVA} &= 8.406 \text{ kVA} \\
 \text{with 20\% expansion} &= 8.406 \times 1.2 \\
 &= 10.0872 \text{ kVA} \\
 \text{Total design current} &= \frac{\text{Total load} \times 1000}{\sqrt{3} \times 400} \\
 &= \frac{8.406 \times 1000}{\sqrt{3} \times 400} \\
 &= 12.13 \text{ A}
 \end{aligned}$$

3.4.2 Demand load calculation for right side apartment

Table 13: Right-Side Apartment Demand Load Calculation

Area	Space	Load	No. of Units	KW per Load	Power Factor	KVA	Diversity Factor	Utilization Factor	Total KVA with Factors
Apartment	master Bedroom	AC	1	1.7	0.85	2	0.8	0.9	1.44
		socket	4	0.15	0.85	0.18	0.5	0.85	0.306
	Bedroom 1	AC	1	1.7	0.85	2	0.8	0.9	1.44

		sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
	Bedroom 2	AC	1	1.7	0.85	2	0.8	0.9	1.44
		Sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
	Dining Room and	Sockets	6	0.15	0.85	0.18	0.5	0.85	0.459
	Pantry & kitchen yard	Exhaust Fan	1	0.75	0.9	0.83333	0.9	1	0.75
		Sockets	6	0.5	0.85	0.6	0.5	0.85	1.51
	Balcony	sockets	4	0.15	0.85	0.18	0.5	0.85	0.306
	Toilets	sockets	3	0.15	0.85	0.18	0.5	0.85	0.2295
	All	lighting	22	0.376	0.9	0.41778	0.66	0.85	0.23437
	Total KVA								8.746

Total demand for a aptment in kVA = 8.746 kVA

with 20% expansion = 8.746×1.2

= 10.4952 kVA

Total design current = $\frac{\text{Total load} \times 1000}{\sqrt{3} \times 400}$

= $\frac{8.746 \times 1000}{\sqrt{3} \times 400}$

= 12.45A

Total demand for an apartment floor = 8.746 + 8.406

= 17.152 kVA

3.5 Total Electrical Demand Load

$$\begin{aligned}\text{Demand load for the ground floor} &= 11.46 \text{ kVA} \\ \text{Demand load for the roof floor} &= 16.14 \text{ kVA} \\ \text{Demand load for the apartment floor} &= 17.152 \text{ kVA} \\ \text{Total Electrical Demand Load} &= 11.46 + 16.14 + 3 \times 17.152 \\ &= 79.056 \text{ kVA}\end{aligned}$$

Transformer Requirement

A 100kVA transformer is selected for the apartment building based on the estimated peak demand of 80kVA. This allows the transformer to operate at approximately 80% loading, which is within an efficient and reliable operating range. The selected capacity also helps maintain stable voltage levels at the MDB by minimizing internal voltage drops. Additionally, a future expansion margin of about 30 kVA is available to accommodate possible load growth without requiring immediate system upgrades.

$$\text{Transformer size} = 100\text{kVA}$$

Generator Ratings

$$\begin{aligned}\text{Fire related systems} &= 6 \text{ KVA} \\ \text{Total load of the Ground Floor} &= 11.46 \text{ kVA} \\ \text{Total load of the Roof Floor} &= 16.14 \text{ kVA} \\ \text{All lighting and socket outlets within one bedroom of} &= 0.306 + (4 \times 11) / 0.9 \\ \text{an apartment} &= 0.3548 \text{ kVA} \\ \text{All lighting and socket outlets within one bedroom of} &= 0.3548 \times 6 \\ \text{6 apartment} &= 2.126 \text{ kVA} \\ \text{Two socket outlets in the pantry area of each} &= 0.6 \text{ kVA} \times 2 \times 6 \\ \text{apartment} &= 7.2 \text{ kVA} \\ \text{Total} &= 38.126 \text{ kVA}\end{aligned}$$

With the future expansion and the effect from motor inrush current we select the generator with 45 KVA rating

4 Cable Sizing and Distribution Diagrams

To ensure safety, compliance, and optimal performance, cables are sized according to BS 7671 (IET Wiring Regulations) using the following step-by-step criteria.

$$\text{Design current} = \frac{\text{Estimated load in kVA}}{\sqrt{3} \times 400}$$

The protective device (MCB/MCCB) rating must satisfy the coordination circuit rule.

$$I_b < I_n < I_z$$

where,

I_b = Design Current

I_n = Current Rating for Protective Device

I_z = Calculated Current for a Circuit with Correction Factors

$$I_z = I \times c_a \times c_g \times c_i \times c_d$$

Where,

c_i = Thermal Factor

c_g = Grouping factor

c_a = Ambient temperature

c_d = Ground condition

Specimen Calculation

Transformer to MDB (Main Distribution Board)

Assume that the distance between transformer and MDB is 50 m and the cable is Free air horizontal/ vertical cable tray (method E). furthermore, temperature is 35 C

$$\begin{aligned} \text{Total load} &= 100 \text{ kVA} \\ \text{Design current} &= \frac{\text{Estimated load in kVA}}{\sqrt{3} \times 400} \\ &= \frac{100 \times 10^3}{\sqrt{3} \times 400} \\ &= 144.33 \text{ A} \end{aligned}$$

Assume the Cable is 4C Cu / XLPE / PVC laid under the method E (free air perforated cable tray) touching.

$$c_a = 0.96$$

$$c_g = 1$$

$$c_i = 1$$

$$\text{Selected rating of the protective device (} I_n \text{)} = 160 \text{ A}$$

If we select 35 square mm 4C Cu / XLPE / PVC cable

$$\text{Current carrying capacity before the derating factors} = 158 \text{ A}$$

(Table 4E2A)

$$\begin{aligned} \text{Current carrying capacity after the derating factors} &= 158 \times 0.96 \times 1 \times 1 \\ &= 151.68 \text{ A} \end{aligned}$$

$$151.68 \text{ A} < I_n$$

Therefore, it is not suitable

If we select 50 square mm 4C Cu / XLPE / PVC cable

$$\text{Current carrying capacity before the derating factors} = 192 \text{ A}$$

$$\begin{aligned} \text{Current carrying capacity after the derating factors} &= 192 \times 0.96 \times 1 \times 1 \\ &= 184.32 \text{ A} \end{aligned}$$

$$I_b < I_n < I_z$$

Voltage drop calculation

$$\begin{aligned} \text{Voltage drop (Table 4E2B)} &= \frac{I_b \times \text{Tabulated voltage drop} \times \text{Cable length}}{1000} \\ &= \frac{144.3 \times 0.87 \times 50}{1000} \\ &= 6.277 \\ \text{Voltage drop in \%} &= \frac{6.277}{400} \times 100 \\ &= 1.569\% \end{aligned}$$

According to BS 7671 voltage drop should be less than 5 % so the **50 square mm 4C Cu / XLPE / PVC** cable is selected as the cable from transformer to MDB.

From generator to the MDB

$$\begin{aligned}\text{Total load} &= 45 \text{ kVA} \\ \text{Design current} &= \frac{\text{Estimated load in kVA}}{\sqrt{3} \times 400} \\ &= \frac{45 \times 10^3}{\sqrt{3} \times 400} \\ &= 64.95 \text{ A}\end{aligned}$$

Assume the Cable is 4C Cu / XLPE / PVC laid under the method E (free air vertical perforated cable tray) touching. Cable length is 20 m

$$\begin{aligned}c_a &= 0.96 \\ c_g &= 1 \\ c_i &= 1\end{aligned}$$

$$\text{Selected rating of the protective device (} I_n \text{)} = 80 \text{ A}$$

If we select 16 square mm 4C Cu / XLPE / PVC cable

$$\begin{aligned}\text{Current caring capacity before the derating factors} &= 100 \text{ A} \\ &(\text{Table 4E2A})\end{aligned}$$

$$\begin{aligned}\text{Current caring capacity after the derating factors} &= 100 \times 0.96 \times 1 \times 1 \\ &= 96 \text{ A}\end{aligned}$$

$$I_b < I_n < I_z$$

$$\begin{aligned}\text{Voltage drop (Table 4E2B)} &= \frac{I_b \times \text{Tabulated voltage drop} \times \text{Cable length}}{1000} \\ &= \frac{64.95 \times 2.5 \times 20}{1000} \\ &= 3.2475 \\ \text{Voltage drop in \%} &= \frac{3.247}{400} \times 100 \\ &= 0.812\%\end{aligned}$$

According to BS 7671 voltage drop should be less than 5 % so the **16 square mm 4C Cu / XLPE / PVC** cable is selected as the cable from generator to MDB.

MDB (Main Distribution Board) to Sub distribution Board at ground floor

$$\begin{aligned}\text{Total load} &= 11.46 \text{ kVA} \\ \text{Design current} &= \frac{\text{Estimated load in kVA}}{\sqrt{3} \times 400} \\ &= \frac{11.46 \times 10^3}{\sqrt{3} \times 400} \\ &= 16.54 \text{ A}\end{aligned}$$

$$c_a = 0.96$$

$$c_g = 1$$

$$c_i = 1$$

$$\text{Selected rating of the protective device (} I_n \text{)} = 25 \text{ A}$$

If we select 2.5 square mm 4C Cu / XLPE / PVC
cable,

$$\begin{aligned}\text{Current carrying capacity before the derating factors} &= 32 \text{ A} \\ &\text{(Table 4E2A)}\end{aligned}$$

$$\begin{aligned}\text{Current carrying capacity after the derating factors} &= 32 \times 0.96 \times 1 \times 1 \\ &= 30.72 \text{ A}\end{aligned}$$

Assumed length of the cable = 5m

$$\begin{aligned}\text{Voltage drop (Table 4E2B)} &= \frac{I_b \times \text{Tabulated voltage drop} \times \text{Cable length}}{1000} \\ &= \frac{16.54 \times 16 \times 5}{1000} \\ &= 1.323 \\ \text{Voltage drop in \%} &= \frac{1.323}{400} \times 100 \\ &= 0.33\%\end{aligned}$$

According to BS 7671 voltage drop should be less than 1 % in power circuits so the **2.5 square mm 4C Cu / XLPE / PVC** cable is selected as the cable from MDB to SDB-GF

For the fire requirement,

Cable Laying method: Heavy-Duty Perforated Cable Tray

$$\begin{aligned}\text{Total load} &= 6 \text{ kVA} \\ \text{Design current} &= \frac{\text{Estimated load in kVA}}{\sqrt{3} \times 400} \\ &= \frac{6 \times 10^3}{\sqrt{3} \times 400} \\ &= 8.66 \text{ A}\end{aligned}$$

$$c_a = 0.96$$

$$c_g = 1$$

$$\text{Selected rating of the protective device (} I_n \text{) } = 20 \text{ A}$$

If we select 1.5 square mm 4C Cu / XLPE / PVC cable,

$$\begin{aligned}\text{Current carrying capacity before the derating factors} &= 23 \text{ A} \\ &\text{(Table 4E2A)}\end{aligned}$$

$$\begin{aligned}\text{Current carrying capacity after the derating factors} &= 23 \times 0.96 \times 1 \times 1 \\ &= 22.08 \text{ A}\end{aligned}$$

Assumed length of the cable = 8m

$$\begin{aligned}\text{Voltage drop (Table 4E2B)} &= \frac{I_b \times \text{Tabulated voltage drop} \times \text{Cable length}}{1000} \\ &= \frac{8.66 \times 27 \times 8}{1000} \\ &= 1.87 \\ \text{Voltage drop in \%} &= \frac{1.87}{400} \times 100 \\ &= 0.467\%\end{aligned}$$

According to BS 7671 voltage drop should be less than 1 % in power circuits so the **1.5 square mm 4C Cu / XLPE / PVC** cable is selected as the cable from transformer to Fire -DB

To satisfy the mandatory requirement for separating essential and non-essential power systems, the vertical distribution network utilizes two completely independent, physically isolated structural perforated vertical cable trays (touching) installed within the building's primary electrical riser duct.

- Ladder 1 (Non-Essential Systems - Utility Only): Dedicated exclusively to the distribution of standard grid power to non-essential floor Sub-Distribution Boards (SDBs) and subsequent apartment distribution boards.
- Ladder 2 (Essential Systems - Backup Generator): Dedicated exclusively to routing critical power from the Automatic Transfer Switch (ATS) to the essential floor SDBs, common area infrastructure and emergency services.

Non-Essential Ladder Contained Circuits (N = 3):

1. Feeder Cable to 1st Floor SDB (Non-Essential)
2. Feeder Cable to 2nd Floor SDB (Non-Essential)
3. Feeder Cable to 3rd Floor SDB (Non-Essential)

Essential Ladder Contained Circuits (N= 4):

1. Feeder Cable to 1st Floor SDB (Essential)
2. Feeder Cable to 2nd Floor SDB (Essential)
3. Feeder Cable to 3rd Floor SDB (Essential)
4. Feeder Cable to Roof Floor SDB (Essential Network)

Number of trays or ladders = 2

Cable laying method = Vertical perforated cable tray (touching)

Grouping factor for the cables in non-essential = 0.81

ladder (Table 4C4) N=3

Grouping factor for the cables in essential = 0.76

ladder (Table 4C4) N=4

Design dimensions for Cable Routing

- Typical Floor Height (1st to 3rd): 3.5m per floor
- Horizontal Run (MDB to Riser Base): 5m (Distance from the main switchboard room to the vertical electrical duct)
- Horizontal Run (Riser to Floor SDB): 2m (Distance to clear the duct and terminate inside the floor panel) and as the Installation Allowance Add 10\% to the raw physical length

4.1 Cable calculation from Main distribution Board

Table 14: Cable calculation from the main Distribution board

	Cable From	Cable To	Length (m)	Ca	Cg	connected load (kVA)	Ib design (A)	In rated (A)	Iz before derating (A)	Iz after derating(A)	Cable cross section (mm ²)	voltage drop factor (table 4E2B)	Voltage Drop (mV/A/m)	Calculated Voltage drop (%)	Selected cable
Main Sub Lines (essential)	MDB	SDB -GF	5	0.96	1	11.46	16.54	25	32	30.72	2.5	16	1.323	0.3308	4C Cu/XLPE/PVC 2.5mm ²
	MDB	SDB -F1	12	0.96	0.76	3.12	4.474	6	18	13.13	1	40	2.147	0.5369	4C Cu/XLPE/PVC 1mm ²
	MDB	SDB -F2	16	0.96	0.76	3.12	4.474	6	18	13.132	1	40	2.863	0.7159	4C Cu/XLPE/PVC 1mm ²
	MDB	SDB -F3	20	0.96	0.76	3.12	4.474	6	18	13.132	1	40	3.579	0.894	4C Cu/XLPE/PVC 1mm ²
	MDB	SDB -RF	24	0.96	0.76	16.14	23.29	25	42	30.643	4	10	5.591	1.397	4C Cu/XLPE/PVC 4mm ²
Main Sub Lines(non-essential)	MDB	SDB -F1	12	0.96	0.81	10.86	15.67	20	32	24.883	2.5	16	3.009	0.752	4C Cu/XLPE/PVC 2.5mm ²
	MDB	SDB -F2	16	0.96	0.81	10.86	15.67	20	32	24.883	2.5	16	4.012	1.003	4C Cu/XLPE/PVC 2.5mm ²
	MDB	SDB -F3	20	0.96	0.81	10.86	15.67	20	42	32.659	4	10	3.135	0.783	4C Cu/XLPE/PVC 4mm ²

4.2 Cable calculation from Ground floor sub DB to mini distribution points

The cable laying Method A (Enclosed in Conduit): Standard multi-core cables drawn through PVC conduits embedded in the wall or floor. When calculating grouping factor assume that single phase cables which going to apartment DB and to common DB are enclosed in conduit with touching furthermore three phase cable to the water pump runs separately.

Table 15: Cable calculation from SDB GF

	Cable From	Cable To	Length (m)	Ca	Cg	connected load (kVA)	Ib design (A)	In rated (A)	Iz before derating (A)	Iz after derating(A)	Cable cross section (mm ²)	voltage drop factor (table 4E2B)	Voltage Drop (mV/A/m)	Calculated Voltage drop (%)	Selected Cable
Ground Floor SDB	SDB-GF	DB-APRT	6	0.96	0.87	6.1	8.804	20	26	21.71	2.5	16	0.845	0.211	4 Cu/XLPE/PVC 2.5mm ²
	SDB-GF	DB-Common	4	0.96	0.87	3.91	5.643	20	26	21.71	2.5	12	0.270	0.067	4 Cu/XLPE/PVC 2.5mm ²
	SDB-GF	Water Pump (Duty)	8	0.94	1	1.67	2.410	6	18	17.28	1.5	40	0.771	0.192	4 Cu/PVC/PVC 1.5mm ²
	SDB-GF	Water Pump (Standby)	8	0.94	1	1.67	2.410	6	18	17.28	1.5	40	0.771	0.192	4 Cu/PVC/PVC 1.5mm ²

4.3 Cable calculation from first floor sub DB to mini distribution points

The cable laying Method B (Enclosed in Conduit) and two cables run into apartment DBs for essential and non-essential loads grouping factor is consider as two cables in enclosed conduit. For the non-essential loads in apartment DB draw 3 phase cables from SDB and for the essential loads draw single phase cables from SDB with the correct phase balancing.

Table 16: Cable calculation from first floor sub DB

	Cable From	Cable To	Length (m)	Ca	Cg	connected load (kVA)	Ib design (A)	In rated (A)	Iz before derating (A)	Iz after derating(A)	Cable cross section (mm ²)	voltage drop factor (table 4E2B)	Voltage Drop (mV/A/m)	Calculated Voltage drop (%)	Selected Cable
First Floor to Third Floor	SDB-F1 (non-essential)	DB-APT1 (Left)	3	0.94	0.8	6.852	9.890	20	26	19.55	2.5	16	0.474	0.118	4 Cu/PVC/PVC 2.5 mm ²
	SDB-F1 (non-essential)	DB-APT2 (Right)	2	0.94	0.8	7.192	10.38	20	26	19.55	2.5	16	0.332	0.083	4 Cu/PVC/PVC 2.5 mm ²
	SDB-F1(essential)	DB-APT1 (Left)	3	0.94	0.8	1.554	6.756	10	22	16.54	1.5	31	0.628	0.157	2 Cu/PVC/PVC 2.5 mm ²
	SDB-F1 (essential)	DB-APT2 (Right)	2	0.94	0.8	1.554	6.756	10	22	16.54	1.5	31	0.418	0.104	2 Cu/PVC/PVC 2.5 mm ²
	SDB-F1	common lobby	4	0.94	1	0.176	0.765	6	14.5	13.63	1	46	0.141	0.035	2 Cu/PVC/PVC 1 mm ²

4.4 Cable calculation from roof floor sub DB

The cable laying Method B (Enclosed in Conduit), For the grouping factor assume that 3 cables for the 3-phase loads run in enclosure

Table 17: Cable calculation from Roof floor sub DB

	Cable From	Cable To	Length (m)	Ca	Cg	connected load (kVA)	Ib design (A)	In rated (A)	Iz before derating (A)	Iz after derating(A)	Cable cross section (mm ²)	voltage drop factor (table 4E2B)	Voltage Drop (mV/A/m)	Calculated Voltage drop (%)	Selected cables
Roof Top	SDB- RF	Kitchen E (3P)	8	0.94	0.7	4.44	6.408	10	19.5	12.83	1.5	27	1.384	0.346	4 Cu/PVC/PVC 1.5mm ²
	SDB- RF	Pool Pump (3P)	6	0.94	0.7	3	4.330	10	19.5	12.83	1.5	27	0.701	0.175	4 Cu/PVC/PVC 1.5mm ²
	SDB- RF	Elevator motor(3p)	8	0.94	0.7	6.667	9.622	10	19.5	12.83	1.5	27	2.078	0.519	4 Cu/PVC/PVC 1.5 mm ²
	SDB- RF	Common DB	4	0.94	1	2.809	12.21	20	25	23.5	2.5	19	0.928	0.232	2 Cu/PVC/PVC 2.5 mm ²

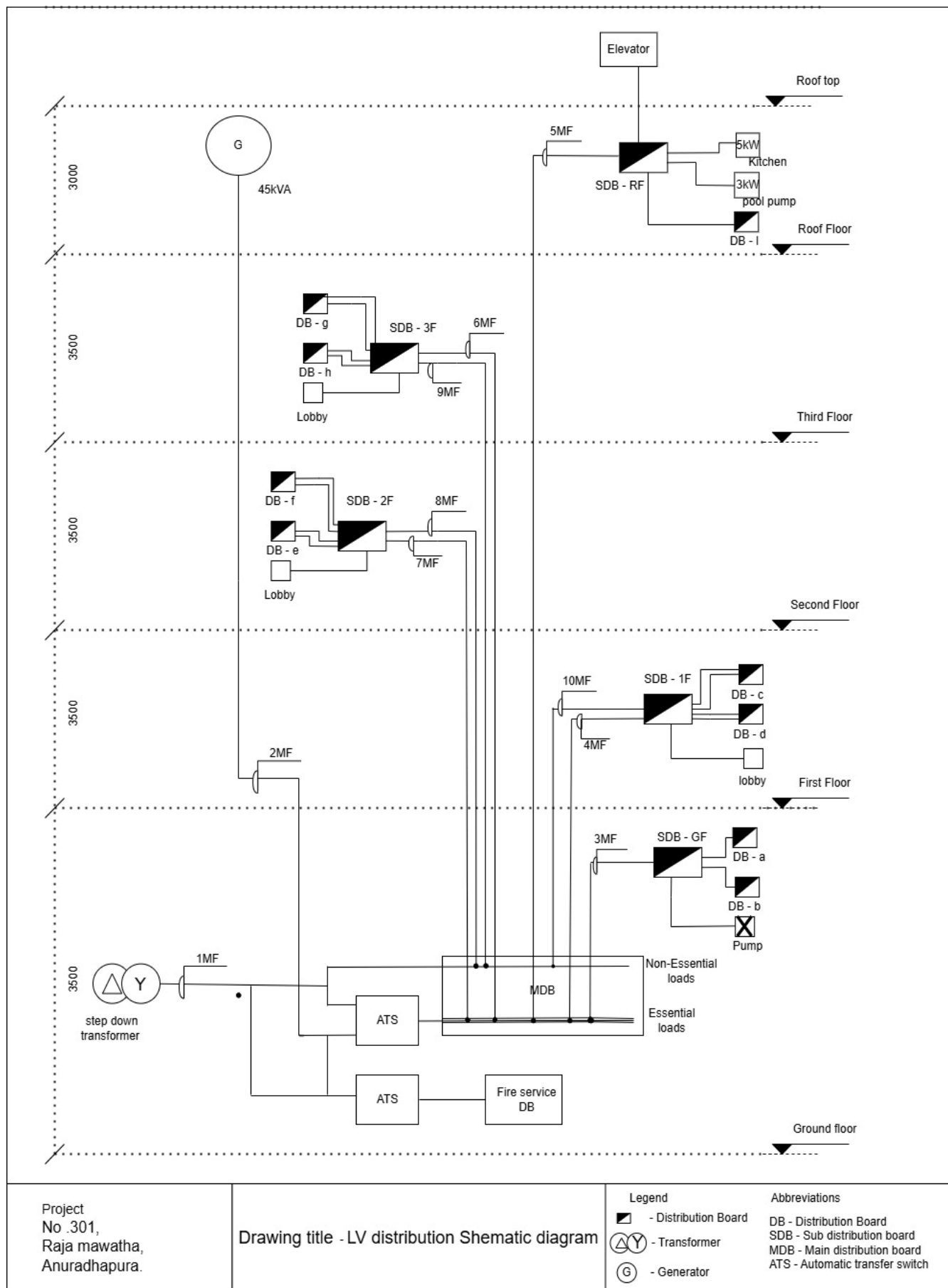


Figure 4: Schematic diagram of the electrical system

Table 18: Main Feeder Cable description

Feeder cable	Cable description
1 MF	4C Cu / XLPE / PVC 50 mm ²
2 MF	4C Cu / XLPE / PVC 16 mm ²
3 MF	4C Cu / XLPE / PVC 2.5 mm ²
4 MF	4C Cu / XLPE / PVC 1 mm ²
5 MF	4C Cu / XLPE / PVC 4 mm ²
6 MF	4C Cu / XLPE / PVC 1 mm ²
7 MF	4C Cu / XLPE / PVC 1 mm ²
8 MF	4C Cu / XLPE / PVC 2.5 mm ²
9 MF	4C Cu / XLPE / PVC 4.5 mm ²
10 MF	4C Cu / XLPE / PVC 2.5 mm ²

Table 19: Sub Feeder Cable description

Cable from	Cable to	Single or three phase	Cable description
Transformer	Fire DB	3-phase	4C Cu / XLPE / PVC 2.5 mm ²
SDB-GF	DB – a (APRT)	3-phase	4C Cu / XLPE / PVC 2.5 mm ²
SDB-GF	DB – b (common)	3-phase	4C Cu / XLPE / PVC 2.5 mm ²
SDB-GF	Pump	3-phase	4C Cu / XLPE / PVC 1.5 mm ²
SDB-1F	DB - c (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-1F	DB - c (non-Essential)	3-phase	4C Cu / PVC / PVC 2.5 mm ²
SDB-1F	DB - d (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-1F	DB – d (non-Essential)	3-phase	4C Cu / PVC / PVC 2.5 mm ²
SDB-1F	Common area lighting	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-2F	DB - e (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-2F	DB - e (non-Essential)	3-phase	4C Cu / PVC / PVC 2.5 mm ²
SDB-2F	DB - f (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-2F	DB – f (non-Essential)	3-phase	4C Cu / PVC / PVC 2.5 mm ²
SDB-2F	Common area lighting	Single phase	2C Cu / PVC / PVC 1 mm ²
SDB-3F	DB - g (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-3F	DB - g (non-Essential)	3-phase	4C Cu / PVC / PVC 2.5 mm ²
SDB-3F	DB - h (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-3F	DB - h (Essential)	Single phase	2C Cu / PVC / PVC 2.5 mm ²
SDB-3F	Common area lighting	Single phase	2C Cu / PVC / PVC 1 mm ²

SDB-RF	DB - i	Single phase	2C Cu / PVC / PVC 1.5 mm2
SDB-RF	Kitchen equipment	3-phase	4C Cu / PVC / PVC 1.5 mm2
SDB-RF	Pool Equipment	3-phase	4C Cu / PVC / PVC 1.5 mm2
SDB-RF	Elevator motor	3-phase	4C Cu / PVC / PVC 2.5 mm2

Table 20: Standard Cable Sizes Selected for General Electrical Loads According to BS 7671 Regulations

General socket	Single phase	2C Cu / PVC / PVC 1.5 mm2
Pantry socket	Single phase	2C Cu / PVC / PVC 2.5 mm2
Lighting	Single phase	2C Cu / PVC / PVC 1.0mm2
Air conditioning	Single phase	2C Cu / PVC / PVC 2.5 mm2

5 Single line diagrams for the electrical system

5.1 Main distribution Board

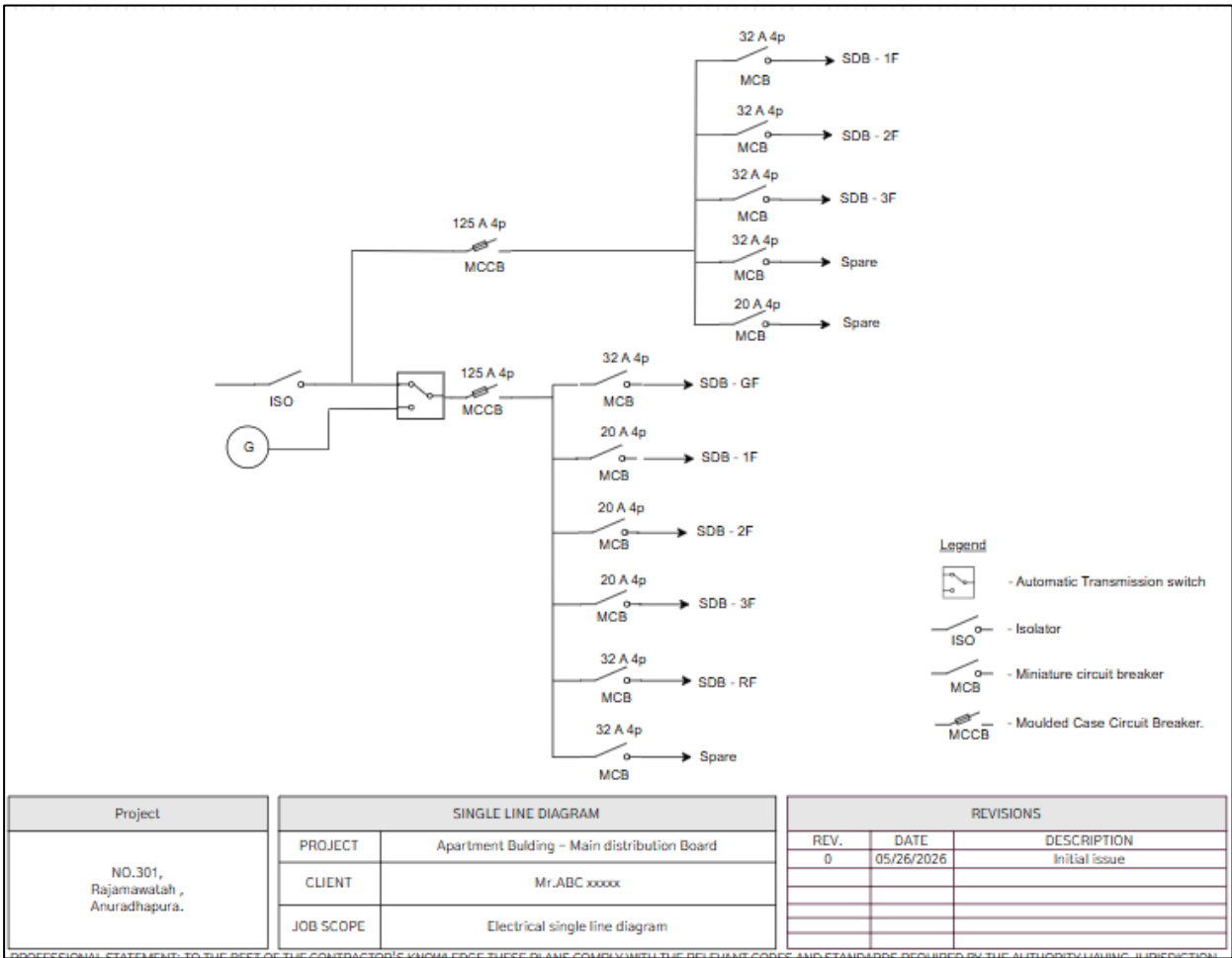


Figure 5: Single line diagram of the Main Distribution Board